Study of Comparators

1 Introduction

In our analog macroscopic world, every physical quantity has a continuous set of values. For example, length, time, speed, and electrical quantities (voltage, current, resistance), etc.

In order to transition from the analog world to the digital, we need to convert the analog quantities into digital ones. The simplest digital signaling system is the binary system, where the magnitude of digital quantities is expressed in the base-2 number system. Here, only two values are required to represent any number: 0 and 1. The binary system can be represented in various ways. For example, electronically, a circuit can represent a digital 1 when current flows and a digital 0 when no current flows. Another possibility is that if any voltage is present, it represents a digital 1, while if no voltage is measured, it represents a digital 0.

When we convert an analog signal to a digital one, one of the fundamental tasks is comparison. In this context, two analog values are compared, but the result of the comparison is a digital value, meaning it can only take two possible outcomes. If the two quantities to be compared are voltages (U_1 and U_2), the two possible outcomes are: $U_1 \geq U_2$ or $U_2 \geq U_1$ (since only two output values are possible, we need to find relationships that mutually exclude each other while covering all possible cases).

In addition to digitization, a very common task in engineering practice is determining whether a quantity exceeds a given value. Such tasks could involve monitoring the water level in a tank, the level of radioactive radiation in a room, or the temperature in a refrigerator or oven. Since the quantities being monitored are usually converted into electrical voltages, this task boils down to a voltage level comparison. Comparators are therefore widely used in practice, so it is worth familiarizing ourselves with them.

For such tasks, a circuit is needed that can accept two analog voltages as inputs and will switch to one of its two stable output states.

Circuits with these properties are called comparators. A comparator can be implemented in various ways. In this lab, we will focus on comparators that can be built using transistors and operational amplifiers.

2 Comparator Circuit with Operational Amplifier

The comparator circuit compares two voltage signals and determines which one is greater. The result of the comparison is reflected in the output voltage level. If the operational amplifier's output switches to the positive supply voltage, then the signal applied to the non-inverting (+) input is greater than the signal at the inverting (-) input. Conversely, if the output of the amplifier switches to the negative supply voltage, the signal at the inverting (-) input is greater, or more positive, than the signal at the non-inverting (+) input.

In the circuit shown in Figure 1, the voltage divider formed by resistors R1 and R2 determines the threshold voltage (U1). If the input voltage (U2) is lower than this, the operational amplifier's output switches to the positive supply voltage (U2 is connected to the inverting input!). If U2 is greater than U1, the output switches to the negative supply voltage.

This configuration is rarely used because it presents several issues. Among other things, the amplifier is highly sensitive to noise due to the high input resistance. Since the input signal is typically noisy, if the average value of the signal is near the threshold voltage, the noise causes it to oscillate between the two states. To avoid this, a better approach would be a circuit where the threshold voltage for switching is different depending on whether the signal

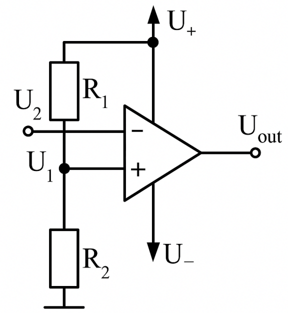

Figure 1: Comparator with operational amplifier

is increasing or decreasing. In other words, the circuit would have hysteresis. This is achieved by the Schmitt-trigger circuit.

3 Schmitt-trigger with operational amplifier

A Schmitt-trigger comparator is a circuit, which has hysteresis. It is shown in Figure 2.

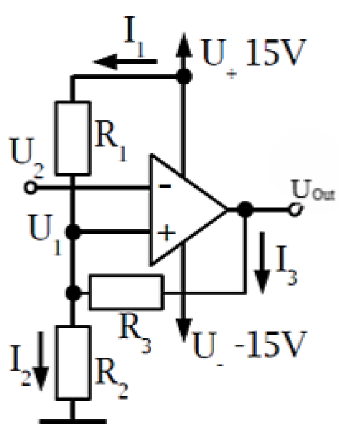

Figure 2: Schmitt-trigger with operational amplifier

In this case, the operational amplifier is powered with +15V and -15V, which represent the ideal output voltage extremes (saturation voltages). The threshold voltage (U_1) is still determined by the U_1 voltage, but now, due to the positive feedback created by the R_3 resistor, it will depend on the output voltage. Since we consider the operational amplifier to be ideal, the U_1 voltage is determined solely by the three resistors and two voltages (U_{Out} and U_+ = 15V):

$$U_1 = R_2 \cdot (I_1 + I_3) = R_2 \cdot (\frac{15 - U_1}{R_1} + \frac{U_{out} - U_1}{R_3})$$

Rearranging the equation for U_1:

$$U_1 = \frac{15}{1 + \frac{R_1}{R_2} + \frac{R_1}{R_3}} + \frac{U_{out}}{1 + \frac{R_3}{R_2} + \frac{R_3}{R_1}}$$

For example, $R_1 = 10k\Omega, R_2 = 10k\Omega, R_3 = 100k\Omega$. Putting these into the equation:

$$U_1 = \frac{15}{2.1} + \frac{U_{out}}{21}$$

By substituting the threshold values $U_{out+} = 15V$ and $U_{out-} = -15V$, we get: $U_1+ = 7.86V$ and $U_1- = 6.43V$.

Therefore, the output voltage of the Schmitt trigger will behave as shown in Figure 3. Thus, for an increasing

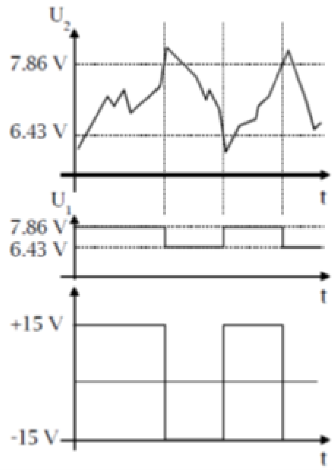

Figure 3: The change in the output of the Schmitt trigger with a varying input signal

input voltage at the U_2 input of the Schmitt trigger, the output voltage will switch from the original positive (+15V) supply voltage to the negative (-15V) supply voltage at 7.86V. For a decreasing input voltage, the output will switch back to the +15V supply voltage at 6.43V.

This also means that the circuit has a hysteresis of $U_H = 7.86V - 6.43V = 1.43V$. If the noise on the input voltage is smaller than this hysteresis, it will not trigger a state change in the circuit.

4 Astable Multivibrator (Relaxation Oscillator) with Operational Amplifier

By modifying the Schmitt trigger circuit, we can create a square wave generator (oscillator) (Figure 4). Similar to the Schmitt trigger, this circuit also takes advantage of the output voltage dependency on the R_2-R_3 voltage divider. Since this is an oscillator, its operation does not require an input signal!

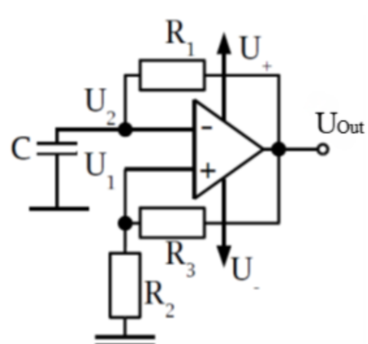

Figure 4: Relaxation Oscillator with Operational Amplifier

The R2 and R3 resistors determine the switching points of the circuit.

$$U_{1s1} = U_+ \cdot \frac{R_2}{R_2 + R_3}$$

$$U_{1s2} = U_- \cdot \frac{R_2}{R_2 + R_3}$$

These are obviously between the supply voltages, U_+ and U_-. Let's assume that the operational amplifier has just switched to the positive supply voltage. This means that $U_2 < U_1$, as the output of the amplifier settles at U_+. At this point, the capacitor C has accumulated a charge of $U_{1s2} \cdot C$, and it starts discharging through the resistor R_1 towards $U_{out} = U_+$. The equation for the charging of a capacitor with an initial charge is:

$$q = q_0 \cdot e^{(-\frac{t}{R \cdot C})} + C \cdot U \cdot (1 - e^{(-\frac{t}{R \cdot C})})$$

In our case $U = U_{out+}$, and $q_0 = U_{1s2} \cdot C$, so the equation is:

$$q = U_{out+} \cdot C \cdot (1 - e^{(-\frac{t}{R_1 \cdot C})}) + U_{1s2} \cdot C \cdot e^{(-\frac{t}{R_1 \cdot C})}$$

The time variation of U_2 is as follows:

$$U_2 = U_{out+} \cdot (1 - e^{(-\frac{t}{R_1 \cdot C})}) + U_{s2} \cdot e^{(-\frac{t}{R_1 \cdot C})}$$

It can be seen that U_2 is exponential (Figure 5), and its time constant is given by the product $R_1 \cdot C$. The charging time of C applies only to one half-period, so $t = T/2$!

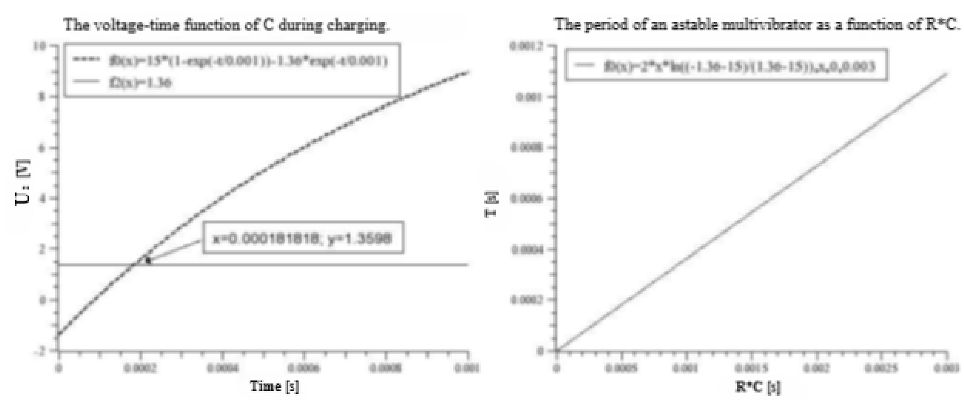

Figure 5: The voltage-time function of the capacitor during charging and the oscillator's T-RC dependence

The continuously increasing U_2 will reach the U_{1s1} switching voltage (the point marked with an arrow in Figure 5), and then, due to the positive feedback, the circuit will switch to its other stable state, i.e., $U_{out} = U_-$, and accordingly, $U_1 = U_{s2}$ (Figure 6).

The time until the switching can be calculated from the following equation:

$$U_{1s1} = U_+ \cdot (1 - e^{(-\frac{T}{2 \cdot R_1 \cdot C})}) + U_{1s2} \cdot e^{(-\frac{T}{2 \cdot R_1 \cdot C})}$$

The equation solved for T gives the period of the relaxation oscillator output:

$$T = 2 \cdot R_1 \cdot C \cdot ln(\frac{U_+ + U_{1s1}}{U_+ + U_{1s2}})$$

After the switching, the voltage across the capacitor cannot change abruptly, so at the moment immediately after switching, $U_2 = U_{s1}$. From this point onward, however, the capacitor begins to discharge, and the voltage at point U_2 exponentially approaches the "new" output voltage U_{out}, i.e., U_-, until it reaches the threshold voltage U_{s2}. At this point, the process has returned to the initial state, and the cycle restarts.

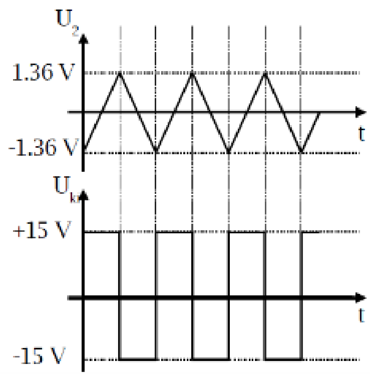

Figure 6: The voltage waveforms of U_2 and U_{out}

5 Transistor Schmitt Trigger

Figure 7 shows a Schmitt trigger implemented with transistors. Due to positive feedback, the circuit has two stable states: either one transistor is conducting, or the other is.

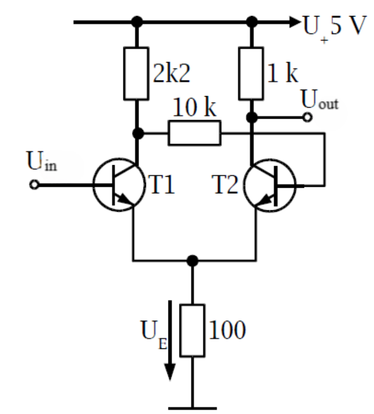

Figure 7: Transistor Schmitt Trigger Circuit

The hysteresis is provided by resistors of different values connected to the individual collectors. Since their values are an order of magnitude larger than the resistor in the emitter, they set the current flowing through the transistors (and thus through the emitter resistor). It is important that if U_{in} is applied at the base of transistor T_1, then the resistor connected to the collector of T_1 should be the larger one.

The voltage on the emitter resistor determines the transistor's turn-on voltage (which is approximately U_E plus about 0.6 V, corresponding to the emitter-base diode voltage). Since both transistors operate in saturation mode (i.e., $U_{CE} \approx 0\,V$), the switching levels can be easily calculated based on the resistor values.

6 Exercises

Power supply should be symmetric ±5V relative to the virtual ground (virtual GND). The virtual ground must also be connected to the circuit!

M1: Build the comparator circuit shown in Figure 1 using a μA741 or TL071 operational amplifier. Use resistors $R_1 = 22\,k\Omega$, $R_2 = 10\,k\Omega$. Slowly vary the input voltage near the switching threshold and determine, using a multimeter, the input voltage at which the output switches state. Is there a difference between the output voltage levels? Verify the switching voltage by calculation in your report.

M2: Modify the circuit according to Figure 2 by adding $R_3 = 100\,k\Omega$. How does the switching threshold change? Verify the measured data with calculations in your report.

M3: Build the relaxation oscillator based on Figure 4 with $R_1 = 10\,k\Omega$, $R_2 = 10\,k\Omega$, and $R_3 = 22\,k\Omega$. Determine the capacitance C so that the oscillator oscillates at 12 kHz. Use the TL071 operational amplifier. Check the waveform shapes at the inputs and output of the operational amplifier. After observing the output waveform, replace the op-amp with a μA741. What changes do you observe in the waveform? What could be the cause of these differences? (Explain in the report based on a comparison of datasheets.) Include pictures of the waveforms generated with both operational amplifiers in your report.

M4: Build the circuit shown in Figure 7 using transistors $T_1 = T_2 = BC182$. Use a +5V power supply relative to ground. Measure the input switching thresholds and output voltage levels of the circuit. Verify the measured data by calculations and explain the output voltage values observed in your report.

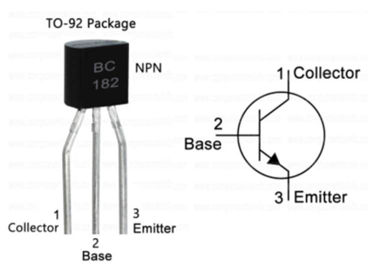

Figure 8: Pinouts for BC182

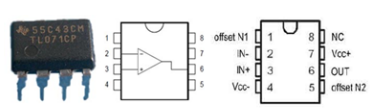

Figure 9: Pinouts for μA741 and TL071

Please include in the report a detailed list of all measuring instruments, generators, cables, and components used, specifying their types, model numbers, quantities, and values whenever available on the devices. Ensure that the photo sizes are optimized so that the report does not exceed 10 MB in total.

The report filename should follow the format: M3_LastName1_LastName2.pdf